LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 7: Lists: A Variable That Holds Many Things

Module 3 | 45 minutes | Lists | Grades 3–5 · ages 9–11

Objective: Create a list, add items to it, read an item by its position, and pick a random item.

Success criteria

- I can create a list and add items.
- I can read an item by position.
- I can pick a random item from a list.

Key vocabulary

- **list**: A variable that holds many items in order.
- **item**: One thing in a list.
- **index**: The position number of an item (starting at 1).
- **length**: How many items are in the list.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Numbered cups (5 min): five cups with numbered labels hold paper slips; learners fetch “item 3”.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	The Fortune Owl: learners follow the steps.
14 min	**Perform**	Code Lab: fortune owl, Item by position
9 min	**Probe**	Code Lab: shopping list, activity
4 min	**Wrap-up**	Exit ticket: Which block tells you how many items a list has?

Unplugged warm-up

Numbered cups (5 min): five cups with numbered labels hold paper slips; learners fetch “item 3”.

Teaching the idea

Lists are ordered collections indexed from 1. Key operations: add, delete all, item n, length. Lists persist between runs in Scratch, so reset them on the green flag. Random selection uses pick random 1 to length.

Analogy: A shopping list: numbered lines, as many as you need.

Live demo: Run the fortune program twice without delete all; show the list doubling. Add delete all.

Questions to ask

Ask	Listen for
What number is the first item?	1.
How do we choose a random item?	pick random 1 to length of the list.
Why delete all at the start?	Lists keep items between runs.

Differentiation

Support
Provide the filled list; learners write the click script.

Stretch
Add a “don’t repeat” rule: delete the fortune after saying it.

Common misconceptions

Misconception	How to fix it
Lists start at item 0	Scratch lists start at 1.
Lists empty themselves when you stop	They keep items until deleted.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Which block tells you how many items a list has?

Homework
Write a list of 5 things a game could store in a list (e.g. high scores, words).

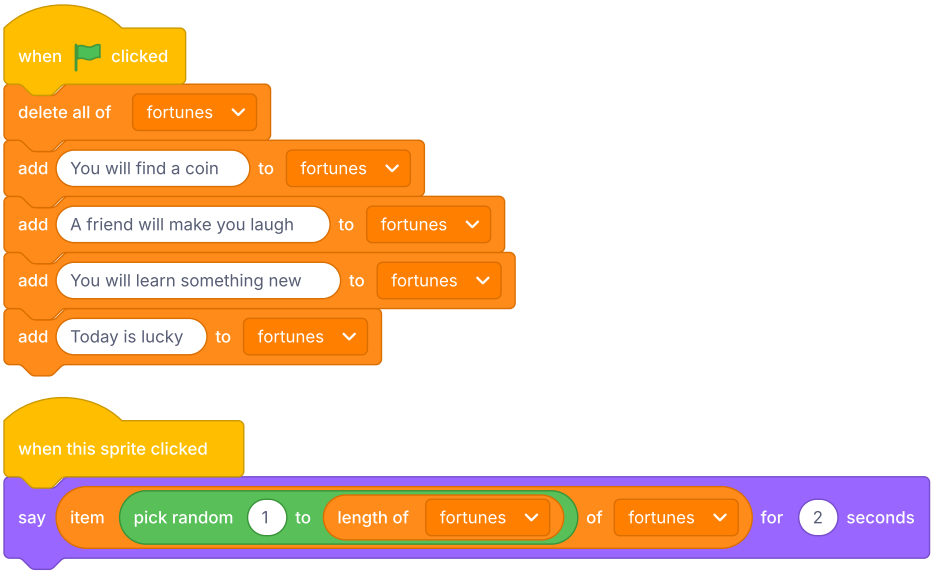
Answer key — Kitabash activities

1. Code Lab: fortune owl (perform · code lab)

Goals checked: The list has 3+ fortunes · Reads an item · Picks at random · Says a fortune when clicked

One solution:

Owl



2. Item by position (perform · multiple choice)

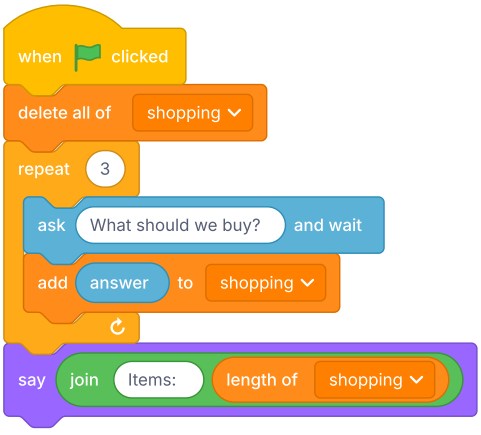
Answer: blue

3. Code Lab: shopping list (probe · code lab)

Goals checked: Asks 3 times · Adds each answer · Starts empty · Says how many

One solution:

Chef



4. Explain your thinking (probe · free response)

Look for: list, many, items. **Example:** A list keeps all the names together, and you can add as many items as you want without making new variables. A high-score table uses a list so it can show every score in order.

Answer key — worksheet

1. **Key words:** list = A variable that holds many items in order. · item = One thing in a list. · index = The position number of an item (starting at 1). · length = How many items are in the list.
2. **Plan your code:** Goals: The list has 3+ fortunes; Reads an item; Picks at random; Says a fortune when clicked.
3. **Item by position:** A. blue — Counting from 1: red (1), green (2), blue (3).
4. **Plan your code:** Goals: Asks 3 times; Adds each answer; Starts empty; Says how many.
5. **Explain your thinking:** A list keeps all the names together, and you can add as many items as you want without making new variables. A high-score table uses a list so it can show every score in order.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.